Task Image Classification Using Cifar-10

Part 1:

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Link Task intro to ML:
https://colab.research.google.com/drive/1yOAcYawtn6h226LSDpUmMjuQO60qLE8z?authuser=0#scrollTo=L4UHusZuK4ol

Part 2 (Answering Questions):

1. Given the following input points ($X1=2$, $X2=4$), ($X1=4$, $X2=2$) and ($X1=2$, $X2=2$) predict their class label.

If Z is 0 or positive, it predicts class 1; if Z is negative, it predicts class 0.

- Final Settings: $W1=2$, $W2=-1$, $b=0.5$.
- Decision Rule: $Z=2X1-X2+0.5$

Prediction for Input Points:

The perceptron predicts **Class 1** for all three points because their calculated Z value is ≥ 0:

- $(2,4) \Longrightarrow Z=2(2)-4+0.5=0.5$ (Class 1)
- $(4,2) \Longrightarrow Z=2(4)-2+0.5=6.5$ (Class 1)
- $(2,2) \Longrightarrow Z=2(2)-2+0.5=2.5$ (Class 1)

- **Draw the decision boundary defined by the final weights and bias.**

  The boundary is the line where $Z=0$:

  - $X2=2X1+0.5$

- **What are the limitations of the perceptron algorithm?**

The main limitation is that the perceptron can only solve problems where the classes can be separated by **a single straight line** (linearly separable data). It cannot solve complex problems like the XOR gate.

2. For Neural Network How many neurons? How many hidden layers? How many learnable parameters in total?

- Total Neurons: Count all the circles: 14 neurons (4 Input + 4 Hidden 1 + 4 Hidden 2 + 2 Output).
- Hidden Layers: The layers between the input and output: 2 hidden layers.
- Total Learnable Parameters (Weights + Biases):
  - Input to Hidden 1: $(4 \times 4 \text{ Weights}) + 4 \text{ Biases} = 20$
  - Hidden 1 to Hidden 2: $(4 \times 4 \text{ Weights}) + 4 \text{ Biases} = 20$
  - Hidden 2 to Output: $(4 \times 2 \text{ Weights}) + 2 \text{ Biases} = 10$
  - Total Parameters: $20+20+10=50$

3. Consider the following 1-d array [3, 4, 6, 3, 2, 3, 4] apply convolution with the following 1-d filter [1, 2, -1] and 0 padding.

Convolution is a sliding-window multiplication and summation. The output is calculated by sliding the 3-element filter across the 7-element array, which results in $7-3+1=5$ output elements:

- $(-1\cdot3) + (2\cdot4) + (1\cdot6) = 11$
- $(-1\cdot4) + (2\cdot6) + (1\cdot3) = 11$
- $(-1\cdot6) + (2\cdot3) + (1\cdot2) = 2$
- $(-1\cdot3) + (2\cdot2) + (1\cdot3) = 4$
- $(-1\cdot2) + (2\cdot3) + (1\cdot4) = 8$ **Output Array:** $[11,11,2,4,8]$

4. For a 1-d array with a length of *L* and a filter of size *F*, what is the length of the output array? How much padding would be required to maintain a constant output size? Explain your answer.

- Output Length (No Padding P =0, Stride S= 1): The output length is always the Input Length (L) minus the Filter Size (F) plus 1.

$$L_{out}=L-F+1$$

Padding to Keep Output Size Constant (Lout=L): To keep the output the same size as the input, you need to add padding (P) to *each side* of the input array.

$$P=2F-1$$

Explanation: This formula ensures that the filter can be centered over every element of the original input array, preventing the output from shrinking.

5. Convolution on the 4x4 array with the 2x2 filter and a stride of 1. The output is a 3x3 matrix.

- **Array:** L=4 [[3, 5, 2, 2],
  [1, -1, 3, -1],
  [0, 6, 3, 7],
  [10, -3, 5, 6]]

- **Filter: F=2**
  [[2, 1],
  [-1, 2]]

The output size will be L−F+1=4−2+1=3. The result will be a 3x3 matrix.

Output Position	Sub-Array Window	Filter	Calculation	Output Value
(1, 1)	(3 1 5 1)	(2 −1 1 2)	(3×2)+(5×1)+(1×−1)+(1×2)=6+5−1+2=12	12

And so on
Output 2D Array:

[12 17 1]
[15 5 16]
[-10 28 20]